

EKRAMUL ALAM

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SUMMARY

CSE graduate with competitive programming experience (Max Codeforces Rating: 1257) and strong foundations in Data Structures & Algorithms, Computer Networks, and Operating Systems. Proficient in C++, Python, Java, JavaScript, and PHP, with hands-on full-stack development across MERN and Laravel ecosystems. Skilled in RESTful API design, relational databases (MySQL, PostgreSQL) and NoSQL (MongoDB), with clean OOP practices and design patterns. Familiar with AI-integrated development using LangChain and FastAPI. Known for clean coding, quick adaptability, and strong problem-solving ability, with the capacity to communicate technical concepts clearly and effectively.

EDUCATION

Khulna University of Engineering & Technology
Bachelor of Science in Computer Science and Engineering

Khulna, Bangladesh
Mar 2022 – Jul 2026

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript, PHP

Core CS: Data Structures & Algorithms, Operating Systems, Computer Networks, OOP, DBMS

Web & Frameworks: React.js, Next.js, Node.js, Express.js, FastAPI, Laravel 10

Databases: MySQL, PostgreSQL (Supabase), MongoDB, Eloquent ORM, Mongoose

AI/ML: LangChain, Vector Embeddings, Prompt Engineering

Tools & DevOps: Git, Linux/Bash, CI/CD Pipelines, Cloudinary, JWT, REST APIs

Communication Skills: English (Full professional and Academic proficiency), Bengali (Native proficiency)

EXPERIENCE

Full-Stack Web Developer (Part-time)

Aug 2025 – Present

TechnoPLUS IT – Remote (Aus)

- Built and maintained a Laravel 9 + Vue.js LMS using Porto/Apiato modular architecture, applying OOP and design patterns (Repository, Service, Observer) across 15+ domain containers
- Independently researched and implemented a custom H5P interactive content integration from scratch, and designed RESTful APIs with role-based access control
- Implemented relational database tables with ordered pivot tables, SoftDeletes, and JSON metadata columns to support a flexible multi-content-type lesson structure

PROJECT EXPERIENCE

HTTP Request Prediction for Faster Web Response using OMNeT++ (C++)

Oct 2025

- Simulated a multi-client HTTP caching system with a first-order Markov-based prediction engine that learns page transition probabilities and pre-caches high-confidence next pages above a configurable threshold
- Built cache lifecycle management with TTL-based expiry, LRU eviction, and periodic cleanup; validated performance across 9 simulation configs (Baseline, Predictive, Sweep, HighLoad, etc.) with server-side latency and hit-rate metrics

AI-Powered Learning Platform (Next.js, FastAPI, LangChain)

Jan 2026

- Developed a university course assistant with AI-powered tutoring and intelligent content search using LangChain and vector embeddings
- Engineered a FastAPI backend with Supabase (PostgreSQL) and row-level security, supporting real-time updates and separate role-based portals for students and instructors

Automated Healthcare System (MERN Stack)

Nov 2024

- Built a full-stack healthcare platform with React.js, Node.js, Express, and MongoDB featuring doctor appointment booking, pharmacy cart, blood donor search, and emergency ambulance services
- Implemented JWT-based authentication with bcrypt password hashing and role-based access control (Admin, Moderator, User) with protected route middleware
- Integrated Cloudinary for image storage and Nodemailer for transactional emails, following RESTful API design with a modular MVC backend structure

MemeGrove – Social Media Platform (Laravel 10, MySQL)

May 2024

- Designed and built a full-featured social platform from scratch with user authentication, image upload pipeline using Intervention Image, likes, comments, following system, and real-time event-driven notifications
- Implemented a custom trending algorithm combining engagement metrics with time decay, alongside tag analytics, infinite scroll pagination, and an admin moderation dashboard

ACTIVITIES

BUET CSE Fest 2026 Hackathon – Finalist | Dhaka, Bangladesh | Jan 2026

BITFEST 2025 – Executive Member | Khulna, Bangladesh | Dec 2024 – Jan 2025

SGIPC (Special Group of Interest in Programming Contest) – Campaign Editor | Khulna, Bangladesh | Aug 2023 – Present

Intra KUET Programming Contest 2023 – Best Team 2k20 Award Winner | Khulna, Bangladesh | 2024

Inter-University Programming Contests (IUPC) – Participant | KUET | 2025